

Operating Systems

2520030042

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SEC-5

WEEK-5

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code

```
Ashish@Ashish: ~
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

int main() {
    int pipefd[2];
    pid_t pid;
    char buffer[100];
    const char *message = "Data from producer";
    struct timespec start, end;

    if (pipe(pipefd) == -1) {
        perror("Pipe failed");
        return 1;
    }

    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        return 1;
    }

    if (pid == 0) {
        close(pipefd[1]);

        read(pipefd[0], buffer, sizeof(buffer));
        printf("Consumer received: %s\n", buffer);

        close(pipefd[0]);
    }
    else {
        close(pipefd[0]);

        clock_gettime(CLOCK_MONOTONIC, &start);

        write(pipefd[1], message, strlen(message) + 1);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double elapsed = (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        printf("Producer sent data.\n");
        printf("Communication time: %.9f seconds\n", elapsed);

        close(pipefd[1]);
        wait(NULL);
    }

    return 0;
}
```

output

```
Ashish@Ashish:~$ nano grep.c
Ashish@Ashish:~$ gcc grep.c
Ashish@Ashish:~$ ./a.out
Producer sent data.
Communication time: 0.000003500 seconds
Consumer received: Data from producer
```